A Controlled Scaling Ladder for Ukrainian Language Models: Registered Predictions and a 124M Baseline

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Abstract

This paper is technical report for training from scratch small-scale decoder-only language model (124M parameters, ~100M non embeddings parameters) in Ukrainian. Ukrainian language has significantly less available data on the internet which makes it harder to train native model from scratch. Data (2.5B-tokens) was taken from a mixture of two high quality sources (fineweb2 and ukr-wiki), data was analysed and additionally cleaned up and normalized for better results (though decisions here were mainly unmeasured). Study ran ladder of 4 different sizes below target to select architecture, initialization and learning rates, predict loss at target. Initial fitting was mostly on 20x token-per-parameter training runs (following Chinchilla's paper 20x receipe), then extended up to 40x and 80x with aim to aditionally fit scaling laws (stopped on 80x assuming that 4 epochs is mostly free on quality). Based on that predictions for a 124M-parameter model at three training horizons were made before training started. The measured losses fell below the predictions by 0.3%, 1.1% and 1.6% at 20, 40 and 80 tokens per parameter. Trained to 80x tokens per parameter — 3.94 expected views of every corpus token — the model reaches loss of 0.9549 bits per character, with horizon gains within 0.003 bpc of a power-law continuation (a consistency check, not a measurement of repetition cost). On the Brown-Ukrainian corpus under the previous Ukrainian from-scratch LLM baseline's protocol, the 124M model reaches 1.394 bpc against 1.467 for the 774M uk4b Large checkpoint scored on the same sentences with its own code. Mid-study evaluation was replaced from a 32-window evaluation subset with the full 6,097-window validation set which did not affect some result (paired first-order effects, moved ≤ 0.0005) and did change others (every fitted curvature, and a learning-rate–horizon confound in the ladder's own longer branches). Across sizes, block-diagonal document masking lowered loss by 0.003–0.006 bpc, RoPE position reset did not produce noticable effect, and residual init suprisingly had negative effect for our rang. Finally under proposed architecture report closes with registered predictions for 500M and 1B runs.

1 Introduction

Ukrainian has limited available training data online. Largets prepared dataset is roughly 60B tokens of public text (Kobza; Haltiuk & Smywiński-Pohl 2025, counted in their tokenizer) against ~15T for English (FineWeb; Penedo et al. 2024), and I was not able to find single-language from-scratch scaling study. The most recent from-scratch Ukrainian generative baseline I found (uk4b; Kyrylov & Chaplynskyi 2023) is a GPT-2 reproduction at 123M–774M parameters trained on UberText 2.0 (Chaplynskyi 2023). Later Ukrainian models (MamayLM, Lapa) are warm starts from multilingual checkpoints, and the multilingual scaling-law study of He et al. (2025) covers Ukrainian only inside a five-language Slavic family. So there is a lack of numbers needed to plan a larger Ukrainian run: the loss-vs-size curve, the value of repeated data, the seed noise floor.

This paper reports a controlled scaling study designed to produce those numbers. It provides (1) measurements from four-rung testing model ladder and a 124M target run, tokenizer and frozen configuration, with the target's loss registered before training at three horizons confirmed with measured within 1.6% accuracy (error have the same sign on all evaluations (§4, §6)); (2) a consistency check on repetition at $\lambda \approx 4$ views per token (§4.6); (3) several per-rung recipe effects with paired seeds: document masking, a RoPE reset that is inert under masking, and an accounting of the small-batch penalty under AdamW and its recovery under Muon+AdamW (§5); (4) comparizon wiith the previous

Ukrainian baseline scored on our sentence set with their own tokenizer and code, which our 33.8M rung matches and our 124M model beats by 0.14 bpc (§6); (5) a learning rate fitting within the ladder and predictions for larger runs (§4.4).

Related work. Small-model ladders that predict a larger run follow Bhagia et al. (2024) and Gadre et al. (2024); the fitting pitfalls are those of Besiroglu et al. (2024) and Pearce & Song (2024), whose case for counting embedding parameters we follow; Hoffmann et al. (2022) and Kaplan et al. (2020) supply the forms. The optimal learning rate falls with horizon (Bjorck et al. 2024); Adam's second-moment horizon and the weight-decay timescale belong in tokens when the batch changes (Porian et al. 2024; Marek et al. 2025; Wang & Aitchison 2024; Bergsma et al. 2025; McCandlish et al. 2018 for critical batch size). Repetition at small epoch counts: Muennighoff et al. (2023), Xue et al. (2023); seed variance: van der Wal et al. (2025). The configuration is standard — AdamW (Loshchilov & Hutter 2019), Muon (Jordan et al. 2024; Liu et al. 2025), RMSNorm (Zhang & Sennrich 2019), SwiGLU (Shazeer 2020), QK-normalisation (Henry et al. 2020; Wortsman et al. 2023), RoPE (Su et al. 2021), WSD (Hu et al. 2024; Hägele et al. 2024), FlexAttention (Dong et al. 2024) — with GPT-2 (Radford et al. 2019) the source of the inherited choices we tested. To my knowledge this is the first single-language, from-scratch scaling ladder for Ukrainian.

2 Data and tokenizer

Corpus. Ukrainian Wikipedia and the `ukr_Cyrl` split of FineWeb-2 (Penedo et al. 2025), mixed 20% / 80% by character budget (Wikipedia capped at 20%; in full it would have been ~40% but would be similar style encyclopedic documents) and sampled without replacement to ~10.2B characters, the cap applied at sampling time: 3,293,685 documents, split at document level into train (3,277,217 documents, 2,522.03M tokens) and validation (16,468 documents, 12.49M tokens). Exact-match deduplication on cleaned text found 16 duplicates (0.0005%); FineWeb-2 is MinHash-deduplicated upstream. A character census puts Cyrillic letters at 80.3%, punctuation and whitespace at 17.1%, Latin letters at 1.6%, other scripts at ~0.001% each, and the remaining ≈ 1.0% (digits and other symbols). In a 197k-document sample, 0.19% of documents (375) were below 50% Cyrillic; 374 of them are bilingual Ukrainian pages, none were dropped. The data cleaner is a conservative normaliser (stress marks, apostrophe code points, invisible characters, whitespace, Wikipedia markup residue; see Appendix A); five heavier rules were built and removed because their false positives outnumbered real artefacts (about 10 to 1).

Tokenizer. Byte-level BPE, 32,768 pieces, trained on the training split of the cleaned corpus: digits are context-free (one id for 7 in 7, −7, .7, (7)); apostrophes stay inside words; no piece spans a newline; the 256 single-byte pieces are present as fallback and we had 0.00% usage on real text; maximum piece length 32 bytes (~16 Cyrillic characters). A vocabulary sweep on 12,897 held-out documents (40.4M characters) gave 3.217 / 3.618 / 4.008 / 4.219 characters per token at 8k / 16k / 32k / 49k: gains of 12.5% and 10.8% for the first two doublings and an extrapolated 9.2% for the next; 32k was selected as a main tokenizer (C.9). All bits-per-character (bpc) values are total bits over an evaluation set's tokens divided by its characters, a common unit across tokenizers (the value still depends on model and tokenizer jointly). Under the document protocols (validation set, §6 documents) the bits include the EOS and BOS targets; they were excluded from counting under the sentence protocol used to compare with uk4b (on bruk set). So the validation set has 4.0305 characters per token.

3 Setup

Architecture. Pre-norm decoder with RoPE, RMSNorm, gated SwiGLU MLP (hidden ≈ 8d/3 rounded to a multiple of 64), QK-normalisation, no biases, tied input/output embeddings, initialisation $0.55/\sqrt{\text{fan_in}}$, head dimension 64, context 2,048. Attention is block-diagonal by document (a FlexAttention mask that restarts at every BOS); RoPE positions are not reset at document boundaries (§5.1). Per-layer matrix parameters are $4d^2 + 3dh$. N_total counts the tied embedding, the per-layer matrices and the RMSNorm gains; the QK-norm gains ($2 \cdot L \cdot 64$) are excluded. The target's matrix (non-embedding) count is 99.09M, chosen close to 100M design target — 14 layers rather than the 12 a total-parameter count would suggest, since the tied embedding is 20% of the target and 79% of the smallest rung.

	d	L	h	heads	N_total	matrix params	tokens at 20×	AdamW lr	Muon lr
P1	128	6	320	2	5,326,464	1.130M	106.5M	5.10e-3	8.16e-3

	d	L	h	heads	N_total	matrix params	tokens at 20×	AdamW lr	Muon lr
P2	192	8	512	3	9,833,664	3.539M	196.7M	4.00e-3	5.00e-3
P3	256	10	704	4	16,422,144	8.028M	328.3M	3.00e-3	3.65e-3
P4	384	12	1024	6	33,826,176	21.234M	676.5M	2.14e-3	2.00e-3
target	768	14	2048	12	124,278,528	99.090M	2.486B	1.19e-3 †	0.45e-3 †

Table 1. The ladder. "20×" is 20 tokens per total parameter; the ray in §4 is fit in N_total, so the target is a 3.7× extrapolation (4.7× in matrix parameters). "Matrix params" excludes the embedding and the norm gains (N_total − embedding is 1.132M / 3.542M / 8.034M / 21.243M / 99.113M). Learning rates are those user for the final masked ladder runs (Appendix E: P1 AdamW 4.93e-3, P2 Muon 5.45e-3, P3 Muon 3.63e-3 on the full validation set) † The target's rates were tuned for its 80× horizon (§4.3). P3's N_total was printed without its norm gains in earlier drafts (16,416,768) and its runs were sized from that count (328.3M tokens = 19.99× the corrected value); the registered fit used the count as it stood: correction impact is at noise level

Optimisation. Muon (momentum 0.95, Nesterov, five Newton–Schulz iterations; update scaled by max(1, rows/cols)^0.5 as in Jordan et al. 2024) on the hidden 2D weights, stored fused (QKV; gate and up) so one orthogonalisation covers each fused matrix; AdamW (β = 0.9, 0.95) on the embedding and 1D parameters. Weight decay 0.1 on all 2D tensors including the tied embedding (from nanoGPT), 0 on norms; gradient clipping at 1.0. Warmup–stable–decay: 200 warmup steps, linear decay over the last 20% of the token budget to 0.1× peak. Batch 32,768 tokens per step, from a P1 sweep run with step-defined hyperparameters (§5.2). Learning-rate vertices per rung from parabola fits at 20× (the runs used the nearest sweep point, Table 1); across the four rungs the Muon vertex scales as d^−1.29 and the AdamW vertex as d^−0.78 (four-point log–log fits, two residual degrees of freedom; leave-one-out −1.16 to −1.45 and −0.70 to −0.90).

Evaluation instrument. Loss on the full validation set as 6,097 non-overlapping, aligned 2,048-token tiles (12.49M tokens), every target position counted. Tiles cut documents at arbitrary points, so the first segment of every tile is scored with truncated context; the protocol is identical for every rung and arm, and the set's fingerprint is frozen (same random seed). Training samples random 2,048-token cuts from the concatenated stream, so a token's expected number of views after T training tokens is $\lambda = T / 2.522B$ and coverage is $1 - e^{-\lambda}$.

Statistics. *Seed noise:* the seed sd of the 20× loss across from-scratch runs is 0.00287 at P1 (n = 5) and 0.00109 / 0.00091 / 0.00112 at P2 / P3 / P4 (n = 5 / 5 / 3; pooled 0.00103; P1 vs pooled F = 7.8, two-sided p = 0.008); the pooled sd of masked branch endpoints at P2–P4, σ = 0.00106, is used for planning. *Design sensitivity:* with σ = 0.00106 and independent arms, the smallest effect detectable at ≈ 80% power (t-quantile approximation; 77–80% under the noncentral t) is 0.0046 / 0.0025 / 0.0017 bpc for 3 / 5 / 8 paired seeds; this minimum detectable effect (MDE) was the study's selection heuristic, not a significance threshold. *Analysis:* arms are paired by seed wherever both exist at the same seeds (exceptions are labelled); paired differences carry the sd of within-seed differences (0.0006 for an initialisation-only change, 0.0021–0.0023 for changes to the computation graph), 95% t intervals and descriptive p-values (about fifteen comparisons, no multiplicity correction). A null is an estimate with its interval, not equivalence; "±" is a standard error.

Compute and protocol. One RTX 5090 (32 GB) for every run; the target's 80× trunk took 11.8 GPU-hours and its two cooldown branches 5.1 more; the ladder's 48 branching runs 11.6; the 57 runs behind Tables 3 and 5 therefore 28.5 GPU-hours in all, with the configuration sweeps of Appendix C, the post-review runs (≈ 13.5 GPU-hours: the 20× run at its own rate and its from-scratch control 4.7 each, the Muon-off attempt 4.2) and the probes of Appendix E on top. Muon's overhead is 19.9% of step time at P1 and ~1% at the target (Appendix J). Predictions were written before each run; the configuration was frozen at P1 (with few verification runs at P2); every mismatch discovered later was logged and its effect computed (Appendix K).

How the configuration was chosen. Most non-default choices were measured against an alternative at P1 before freezing, each arm at its own learning-rate vertex, on the earlier 32-window instrument; Table C.1 lists every choice with the alternative, its effect size, interval and seeds, and marks the inherited ones. Three carried measurable weight: QK-normalisation (+0.012 bpc without it at P1 and P2, and twice the seed variance), dropping GPT-2's $1/\sqrt{(2L)}$ residual scaling (+0.005 to +0.010 with it, training loss also worse — an optimisation cost at this depth, this was also checked on P2 with similar results), and Muon on the hidden matrices (§5.3); SwiGLU vs GELU and RMSNorm vs LayerNorm

were below detectable level. Vertices re-measured on the full validation set moved < 5%.

4 The ladder and the target

4.1 Baselines

Causal (unmasked) 20× losses from scratch: P1 1.50484 (sd 0.00287, n = 5), P2 1.37098 (0.00109, 5), P3 1.27492 (0.00091, 5), P4 1.16253 (0.00112, 3). Masked branch endpoints at the same seeds — the arm the target uses — at 20× / 40× / 80×: P1 1.50010 / 1.46526 / 1.44138; P2 1.36550 / 1.33104 / 1.30838; P3 1.27222 / 1.23934 / 1.21866; P4 1.15967 / 1.12957 / 1.11020. At P1 (seeds 47–49) a 20× branch reproduces a from-scratch masked run to +0.0000 / +0.0002 / −0.0002; at P2–P4 the three-seed from-scratch masked runs of §5.1 differ from the five-seed branch means by +0.0009, −0.0003 and +0.0003 (different seed counts; Appendix D).

4.2 The 20-tokens-per-parameter ray

Twenty tokens per parameter is the Chinchilla heuristic (Hoffmann et al. 2022): the one same-compute comparison available is the ray's 0.949 for a 249M model at 20× against 0.955 measured for the 124M model at 80× (§4.3). at the same budget it favours 20× over 80× at this compute. The ray is a controlled slice through the (N, D) plane. A three-parameter power law fit to the masked 20× endpoints:

$$L(N) = 0.6803 + 73.16 \cdot N_{total}^{-0.290},$$

reproduces the four rungs with residuals ≤ 0.0009 and predicts 1.0091 at the target; this is the fit that was registered before the run (its script and inputs are released with the prediction file, Appendix K). An unweighted least-squares fit on the same means gives $0.6861 + 75.6 \cdot N^{-0.293}$ and prediction of 1.0098 at target, so the estimator moves the prediction by less than the sensitivity below. With four points and three parameters the uncertainty is a sensitivity analysis: dropping P1, P2, P3 or P4 moves the target prediction to 1.0022, 1.0065, 1.0120 and 1.0266.

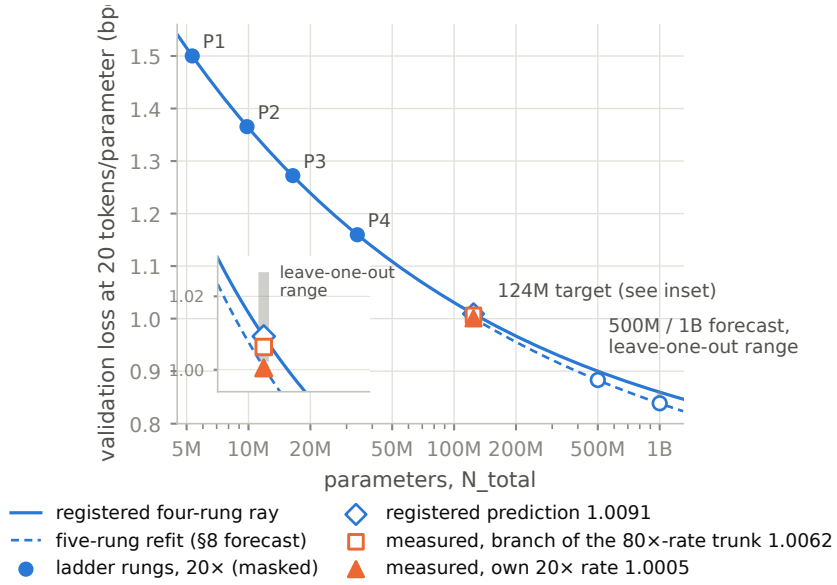


Figure 1. The 20-tokens-per-parameter ray. Masked 20× losses of the four rungs (seed means; seed sd ≤ 0.0015) with the four-rung ray registered before the target run, $0.6803 + 73.16 \cdot N^{-0.290}$, its 124M prediction 1.0091 and the leave-one-out range [1.0022, 1.0266] (inset). The target's two measured 20× points: the cooldown branch of the trunk tuned for 80× (1.0062, −0.3%) and the full run at the horizon's own learning rate (1.0005, −0.85%). Dashed: the five-rung refit on the own-rate point with its 500M and 1B forecasts and their leave-one-out ranges (§8).

4.3 The target

The target was trained once (seed 47) at 32,768 tokens per step for 9.94B tokens (80 tokens per total parameter, λ = 3.94, 98.1% coverage), with cooldown branches at 20× and 40× (of total tokens) from the same run: a branch copied the stable-phase trunk at 80% of a horizon and runs the 20% linear decay to that horizon, so we got losses at several horizons at the same peak learning rate (Appendix F). After that, the 20× horizon was trained again from scratch at its own learning rate (Muon 0.85e-3, the ladder's width-transferred 20× vertex; AdamW unchanged; same seed, tokens and schedule shape), the measurement the first draft had replaced with a model-based correction.

horizon	tokens	λ	registered	measured, branch of the 80×-rate trunk	error	at the horizon's own rate
20×	2.49B	0.99	1.0091	1.0062 (1.0061 from scratch at the same rate)	−0.0029 (−0.3%)	**1.0005** measured (full run, Muon 0.85e-3): −0.0086 (−0.85%)
40×	4.97B	1.97	0.9858	0.9747	−0.0111 (−1.1%)	model-based: penalty 0.001–0.002 → −0.012 … −0.013
80×	9.94B	3.94	0.9704	0.9549	−0.0155 (−1.6%)	own rate by construction if the horizon exponent at the target is P3's; up to 0.002 detuned if it is P2's (§4.4)

Table 2. Registered predictions against measurements. The 40× and 80× predictions combine the ray with the ladder's horizon extrapolation using the registered inputs $A \approx 0.068$ (from the A(N) trend of §4.4) and $\beta = 0.607$. The last column is a measurement at 20× and a sensitivity analysis at 40× and 80×.

The measured losses are below the registered predictions at every horizon; for the branches of the 80×: −0.3%, −1.1% and −1.6%. The 20× and 40× branches, however, ran at a 80x rate for Muon (0.45e-3), whereas every ladder rung ran at its own 20× vertex; width transfer puts the target's 20× vertex at 0.85e-3 and the horizon rule of §4.4 its 40× vertex at ≈ 0.61e-3. Before the 20× run at its own rate was made, we registered its outcome from a parabola in ln(lr): penalty $a \cdot \ln^2(0.85/0.45) = 0.0057$–$0.0073$ with the curvatures $a = 0.014$–0.018 of the registered analysis, 0.0049–0.0078 across the full span of fitted curvatures (0.0120–0.0192, Appendix E), hence a prediction of 0.999 ± 0.003 and a paired difference of −0.005 to −0.009 against the branch. The run gave **1.0005**: 0.0057 below the branch, inside the registered range at its small end, an implied curvature of 0.014 at the target if 0.85e-3 is the vertex. A control run from scratch at the branch's own rate (0.45e-3, otherwise identical) gave **1.0061** against the branch's 1.0062 — registered 1.0062 ± 0.002 — so branch and trunk agree at the target to 0.0001 at one seed, as at P1, and the 0.0057 is the rate. Against the registered ray the own-rate 20× point is −0.85%, outside the leave-one-out range [1.0022, 1.0266] on the low side, as the sensitivity analysis had said it would be (−0.8 to −1.2%). The like-for-like 20× comparison with the ladder is therefore a measurement, at its own rate, the 124M model beats the four-rung forecast by 0.85% at 20× — and by 1.1% and 1.6% at 40× and 80× as branches, with 40× carrying a model-based correction of 0.001–0.002 that would make it ≈ 1.2–1.3%. A log-linear alternative to the power-law horizon extrapolation predicted 0.865 at 80×; though measurement did not confirm it. An earlier attempt at the same run had Muon switched off by mistake (every parameter on AdamW at 1.19e-3): 1.0316, logged in Appendix K and used in §5.3 as a loose bound.

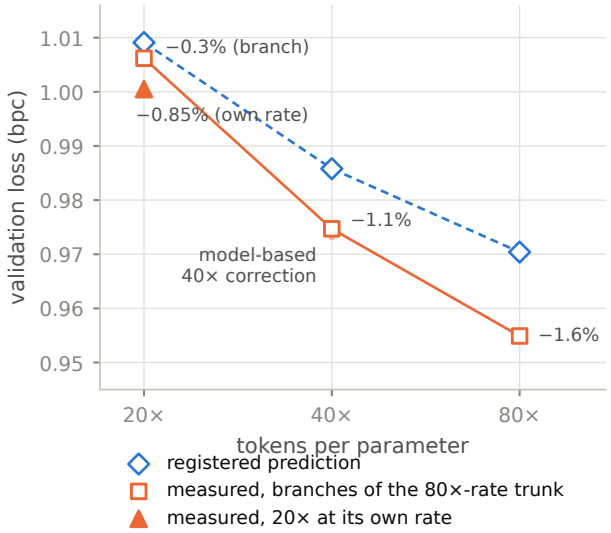


Figure 2a. The target's three horizons: registered predictions (A ≈ 0.068, β = 0.607 on the registered ray) against measurements. Squares: cooldown branches of the run tuned for 80×; triangle: the 20× horizon trained at its own rate. The bar at 40× is the model-based correction for that branch's detuning (0.001–0.002 bpc). The log-linear alternative predicted 0.865 at 80×, off the scale, and is rejected.

4.4 Horizon gains, and a detuning the ladder shares

Along each rung, cooldown branches at 20×, 40× and 80× give three points of L(r) = E + A·r^−β with r = D/(20 N_total), so that A is the loss available beyond the 20× point and is comparable across rungs (Table 3). As measured, A falls with N (0.1102, 0.1024, 0.0890, 0.0844; ∝ N^−0.153), that trend predicted A ≈ 0.068 at the target, and the target's own fit gives 0.081–0.089 with its 20× point at its own rate (0.085 from the three branches as measured): the ladder appeared to under-predict the target's 20×→80× gain by 18% (own-rate 20× point) to 33% (branches as measured).

The target's 80× point is at a rate tuned for 80×; the ladder's 40× and 80× branches hang off trunks run at each rung's 20× vertex, and the optimal rate falls with horizon (Bjorck et al. 2024). The ladder's own single-seed 80× sweeps measure by how much: at P2 the 80× vertex is 4.2e-3 against the 20× vertex 5.5e-3 (horizon exponent γ = −0.20), and running the 80× branch at the trunk's rate (5.0e-3, Table 1) costs 0.0006 bpc (0.0015 had the trunk run at the vertex); at P3 it is 1.9e-3 against 3.65e-3 (γ = −0.46; four-point parabola, residual 1e-4) and the cost is ≈ 0.005; P1 and P4 were not probed. Applying those costs to the 80× endpoints (and the γ-rule to the 40× ones) moves P2's β from 0.605 to ≈ 0.57 and its A from 0.102 to 0.106, and P3's β from 0.65 to ≈ 0.48 and its A from 0.09 to 0.12 — an A that no longer falls with N and a β that no longer rises — and the 18% excess of the target's measured 20×→80× gain (0.0456 at its own 20× rate against the registered 0.0387) is inside what this bias produces. We therefore report the per-rung fits as measured, withdraw "A flattens at the target" as a finding, and report no trend found in β. The horizon exponent is shape-dependent (−0.20 at d = 192, −0.46 at d = 256); the target's 80× rate assumed P3's value, and if P2's applies instead the 80× point is itself detuned by ≈ 0.002 (Table 2). The lesson is the same on both sides: branch learning rates must be set per horizon, on the ladder as much as on the target. The target's β now rests on its 40× point alone: with the 20× point measured at its own rate and the 40× branch corrected at the measured curvature, β = 0.54 (0.52–0.60 across the ladder's curvatures; 0.38 with the 40× branch taken as measured, 0.67 from the three branches as measured). A 0.001 change in the 40× point moves β by ≈ 0.13, so the value is reported as a range.

4.5 β per rung

rung	β	± SE	A
P1	0.546	0.028	0.1102
P2	0.605	0.005	0.1024

rung	β	± SE	A
P3	0.646	0.019	0.0890
P4	0.637	0.021	0.0844
target	0.54 (0.52–0.60 across the 40× correction; 0.38 with the 40× branch as measured; 0.67 from the three branches)	—	0.081–0.089 (0.111; 0.085)

Table 3. Per-rung data exponents from masked cooldown branches at 20×/40×/80× as measured, i.e. with the 40× and 80× branches at each rung's 20× rate (§4.4); five seeds at P1–P3, three at P4; three points and three parameters per seed; ± is the standard error of the seed mean, which absorbs the within-trunk correlation of a seed's three endpoints. P2's SE implies a per-seed sd of 0.010 against 0.036–0.062 at the other rungs: the single P2 run used as the worked example in §7 gives β = 0.568, which an SE of 0.004 would place 3.7 sds below P2's mean.

Taken at face value the rungs differ (inverse-variance $\chi^2 = 11.6$ on 3 df, $p = 0.009$; P1 and P3 against P2's narrow interval; P1 and P3 differ by 0.10, 3.0 SE). The detuning of §4.4 biases β upward by rung-dependent amounts, larger at P3, and P2's interval is unverified. The power-law form itself was validated at P2 with five horizons at ratio √2 and three parameters: $\beta = 0.620$, residuals ≤ 0.0003, and the three-point fit predicted the two held-out horizons to −0.0005 and +0.0001 (Appendix E; subset instrument).

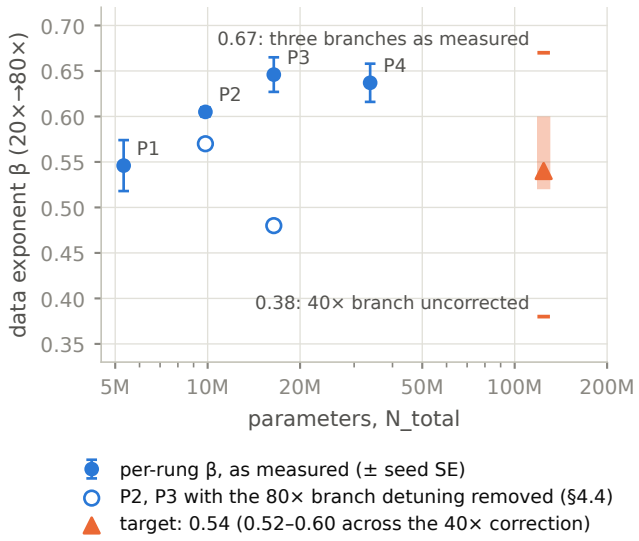


Figure 2b. Per-rung data exponents β from the three masked branches with seed standard errors (Table 3). Open circles: P2 and P3 with the 80× branch's measured detuning removed (§4.4), a sensitivity rather than a measurement. The target's β rests on its 40× branch: 0.54 at the measured curvature, 0.52–0.60 across the ladder's curvatures, 0.38 with the 40× branch uncorrected and 0.67 from the three branches as measured.

4.6 Repetition

At $\lambda = 3.94$ the target's 40× → 80× gain is 0.0198 with the 40× branch as measured and 0.0181–0.0187 after its correction (§4.3). A power law continued from the 20× → 40× gain (0.0258 from the own-rate 20× point; 0.0269–0.0275 with the 40× correction) at $\beta = 0.607$ predicts 0.0169 / 0.0176–0.0180; across $\beta \in [0.50, 0.67]$ and the correction, the measured gain lies between 0.0013 below and 0.0036 above the continuation. The loss kept improving through 3.94 expected views with no flattening beyond the specified extrapolation, to within ≈ 0.004 bpc. This is a consistency check on one model and one corpus, not a measurement of the cost of repetition: the comparison against fresh data at equal compute was not run, and the residual's uncertainty (seed, exponent, correction) is not propagated. Muennighoff et

al. (2023) predict ≈ 0.002 bpc at λ = 4 for a model of this size, below what three seeds at P2 could detect. Views are Poisson-distributed under random cuts (2% of tokens unseen at λ = 3.94, some seen eight times or more) and each repeated view arrives with a different left context; hypothesis is (though not tested) that this is mildly more favourable than epoch shuffling.

5 Ablations across rungs

5.1 Document masking and RoPE reset

Block-diagonal attention by document lowers the 20× loss at every rung. Two estimates exist. (a) Three seeds trained with and without the mask at the frozen configuration, paired: −0.0045 (P2; within-seed sd 0.0014, p = 0.03), −0.0033 (P3; 0.0009, p = 0.02), −0.0030 (P4; one seed); the same arm at P1, run before the learning-rate re-tune, gave −0.0063 (0.0008, p = 0.005). (b) The five-seed masked 20× endpoints against the five-seed causal baselines at the same seeds: −0.0047, −0.0055, −0.0027, −0.0029 (P4: three seeds). Both put the recipe effect between 0.003 and 0.006 bpc, shrinking roughly as N^−0.3 to N^−0.5 (four-point fits −0.34 to −0.41; P2–P4 alone −0.32 to −0.48), ≈ 0.0015 at the target. Masking also trains 1–4% faster. Each arm is evaluated under its own mask.

The mask could help for two reasons: the model trains better without cross-document attention, or the causal model is merely handicapped at evaluation by the unrelated documents in front of each one. The second is testable: score the causal-trained P4 with the mask switched on. If the earlier documents were only a distraction, hiding them would help; instead the loss rises by 0.005 bpc, and by the same amount whether 2 or 1,000 foreign tokens are hidden (Appendix F). Lost information would cost more the more is hidden; a fixed cost per document is a train/test mismatch at the document start. The masked arm's advantage is therefore a training effect. The test is one-sided (the masked-trained model has not been scored without its mask).

Resetting RoPE positions at document boundaries on top of the mask measures −0.0001, −0.0004, +0.0007, +0.0001 at P1–P4. The null is expected: rotary scores depend only on the offset j − i (Su et al. 2021), and under a block-diagonal mask every query attends only to keys in its own document, so a per-document constant added to all positions changes no score; QK-normalisation precedes the rotation and preserves the identity. The practical point: because the two are routinely bundled as if the reset were required: under the mask only the mask matters.

5.2 Batch size at fixed tokens

At P1 and fixed tokens, halving the batch to 16,384 tokens per step costs +0.0275 bpc under Muon+AdamW (five paired seeds, unmasked configuration, t = 22; +0.0299 [0.0280, 0.0318] at the masked frozen configuration, three seeds paired with the masked reference) and doubling to 65,536 costs +0.0019 (t = 5.0; +0.0008, t = 1.8, masked), a 15× asymmetry; learning-rate mismatch under the √B rule accounts for ~10% of the halving penalty by the parabola curvatures of both groups. With AdamW on every parameter at the frozen configuration (three paired seeds) the same comparison gives +0.0249 [0.0198, 0.0300] and +0.0009 [−0.0027, +0.0044]: the asymmetry is optimizer-independent, which rules out an orthogonalisation-specific mechanism.

Three of the recipe's settings are defined in steps, so at fixed tokens halving the batch halves each of them in tokens: warmup, the Adam EMA horizons 1/(1−β), and the weight-decay timescale 1/(ηλ). The learning rate is the fourth knob; the √B rule says it should fall by √2. That these belong in tokens is known (Porian et al. 2024; Marek et al. 2025; Wang & Aitchison 2024; Bergsma et al. 2025); Table 4 does the accounting at one rung by putting each back in tokens at the halved batch, one at a time and all together (settings and algebra in Appendix I).

Under AdamW, all four together recover the whole penalty and more: the restored small-batch run ends 0.0028 below the full-batch one (t = 10.9), consistent with 32,768 tokens being at or above P1's critical batch size (McCandlish et al. 2018). One at a time, the EMA horizons recover about half of the penalty, the weight-decay timescale a third, the learning rate a tenth and warmup nothing measurable.

Under Muon+AdamW, the same four changes plus Muon's momentum 0.95 → 0.975 (the same rule applied to its EMA) recover 95% of the penalty with both learning rates scaled by 1/√2 and 91% with Muon's rate left as is. Two things differ from AdamW. Muon+AdamW is not less batch-sensitive to begin with (+0.0299 against +0.0249, about four standard errors apart). And after the restoration the small-batch run only matches the full-batch one (+0.0014, within

noise) or sits 0.003 above it (Muon's rate unscaled); it does not beat it. Per-seed values and intervals in Appendix I and D.3.

knob restored at mbs 8 (AdamW)	recovery	share of penalty [95% CI]	t (df 2)
Adam EMA horizons β = (0.9, 0.95) → (0.95, 0.975)	+0.0122	49% [3%, 95%]	5.6
weight decay 0.1 → 0.05 at fixed η (1/(ηλ) in tokens)	+0.0083	34% [6%, 61%]	4.3
learning rate × 1/√2	+0.0022	9% [3%, 15%]	6.2
warmup 200 → 400 steps	+0.0011	4% [−33%, 42%]	0.5
sum of the four (unrounded)	+0.0237	95%	
all four together (η/√2, λ 0.0707, warmup 400, β .95/.975)	+0.0277	111% [106%, 116%]	22.5

Table 4. Recovery of the halving penalty (+0.0249) at P1 under AdamW, three paired seeds per arm; shares are ratio-of-means with delta-method intervals (the mbs-8 arm is shared by numerator and denominator); EMA and weight decay are not ordered (paired t = 1.1). Per-seed values in Appendix D.3.

The rule that follows: define warmup, the weight-decay timescale and the optimizer EMA horizons in tokens and derive steps per batch (the decay phase, a fraction of the token budget, already is); a batch-size sweep at fixed tokens that leaves them in steps measures the schedule, not the batch. The near-free doubling plausibly has the same explanation in reverse.

5.3 Optimizer and schedule

Muon on the hidden matrices against AdamW on everything, each at its own vertex, is −0.0096 at P1 and −0.0077 at P2 at equal tokens on the earlier instrument, and −0.0170 at the frozen configuration on the full validation set, both arms masked; its 19.9% overhead at P1 buys ≈ 0.010 bpc of tokens there, so the equal-wall-clock advantage at P1 is ≈ 0.007 on the frozen configuration and nil on the earlier one, while the overhead is ~1% at the target. An early WSD-vs-cosine comparison at a shared, untuned rate (WSD better by 0.0793, eight seeds) is retracted; at each schedule's own vertex cosine is nominally better by 0.0062 with one seed each, which neither establishes nor excludes the parity Hägele et al. (2024) report; WSD is kept for its cooldown branching. The remaining configuration comparisons are in Appendix C. A one-seed data point at the target: a 20× run with Muon switched off by mistake — every parameter on AdamW at the embedding rate 1.19e-3, otherwise identical to the own-rate run of §4.3 — scored 1.0316 against 1.0005, a 0.031 bpc gap; the AdamW rate was not re-tuned for the hidden matrices, so this bounds Muon's advantage at 124M only loosely, from below, at an untuned rate.

6 External evaluation: Brown-Ukrainian and uk4b

Corpus and protocols. Brown-Ukrainian (BrUK; Starko & Rysin, `good` split), 805 verified fragments of literary Ukrainian, scored two ways on the same text with the same cleaner (a near no-op there). *Sentences* — uk4b's protocol, reproduced exactly: 36,038 sentences from `tokenize_uk`, 35,575 after exact-duplicate removal, **32,960** after dropping every sentence that occurs verbatim in our training corpus (2,615 = 7.35%); each sentence is wrapped `<|bos|>` … `<|eos|>`, every sentence token is scored from the BOS-prefixed context, the first token included, the end token is not scored, and the score is total bits over total characters (3,490,610). Sentences are packed into 2,048-token windows, never split, which under the block mask is equivalent to scoring them independently (and is valid only for masked models, which all of ours in Table 5 are; uk4b Large is scored by its own code, one sentence at a time). *Documents*: the fragments as a BOS…EOS stream under the tile protocol of §3, EOS and BOS targets counted, with no sentence-level removal (0 of 805 full-document hashes match a training document; 7.35% of the sentences they contain do). A line-level protocol used in earlier drafts was dropped (§7). Filters, hit counts and MinHash checks: Appendix G.

	val (tiles)	BrUK documents	BrUK sentences
P1	1.4414	1.5395 (0.0018)	1.7189 (0.0035)
P2	1.3084	1.4131 (0.0012)	1.6096 (0.0024)
P3	1.2187	1.3271 (0.0007)	1.5372 (0.0022)
P4	1.1102	1.2234 (0.0007)	1.4478 (0.0005)
target	0.9549	1.0756	1.3267
uk4b Large, same sentences			**1.4665**

Table 5. 80× losses in bpc on three instruments (seed sd in parentheses; sentences over five seeds at P1–P3 and three at P4; val seed sd 0.0015 / 0.0008 / 0.0006 / 0.0005 at P1–P4 over five, five, five and three seeds (D.1); the target is a single run). uk4b Large: the final published checkpoint (774M parameters; 21.4B training tokens in their 50k SentencePiece tokenizer over a 4.3B-token corpus, ≈ 5 passes; ≈ 13× our training compute by the 6ND proxy, our cooldown branches excluded), scored with the authors' own code and tokenizer on our 32,960 sentences.

Three observations. (i) *The extrapolation error has the same sign on every instrument.* Four-rung rays fit on the 80× endpoints predict the target at 0.9699 (val), 1.0882 (documents) and 1.3331 (sentences); the measurements are below them by 1.55%, 1.16% and 0.48% — outside the leave-one-out range on the first two ([0.958, 0.995], [1.078, 1.109]), inside it on sentences ([1.318, 1.364]) — and the error shrinks with granularity. These are three protocols on one model and one text, not three replications, and the ladder's 80× endpoints carry the detuning of §4.4. (ii) *Protocol offsets.* Sentence minus document is 0.179 at P1, 0.224 at P4 and 0.251 at the target, growing with N; sentence-level gains per rung are 81–82% of val gains and 78% at the target step, document-level gains 95–96%. Part of the sentence offset is a fixed cold-start cost: the first token of every sentence, predicted from BOS alone, costs ≈ 9.2 bits at every size from 5M to 124M — 0.087 bpc of the sentence score that no model in the range reduces (implied exactly by the two scorings of each checkpoint, with and without the first token). The rest mixes the value of context with different retained targets (the document protocol scores the sentences the sentence protocol removes) and special-token accounting, so these offsets do not estimate how much context the models use; the matched-target evaluation that would is Appendix L.2. (iii) *Documents minus val* is 0.098–0.121 and grows slowly with N (0.118 for the target at 20× against 0.121 at 80×; the target's 20× → 80× document gain is 94% of its val gain, as on the ladder); register, length and composition were not separated.

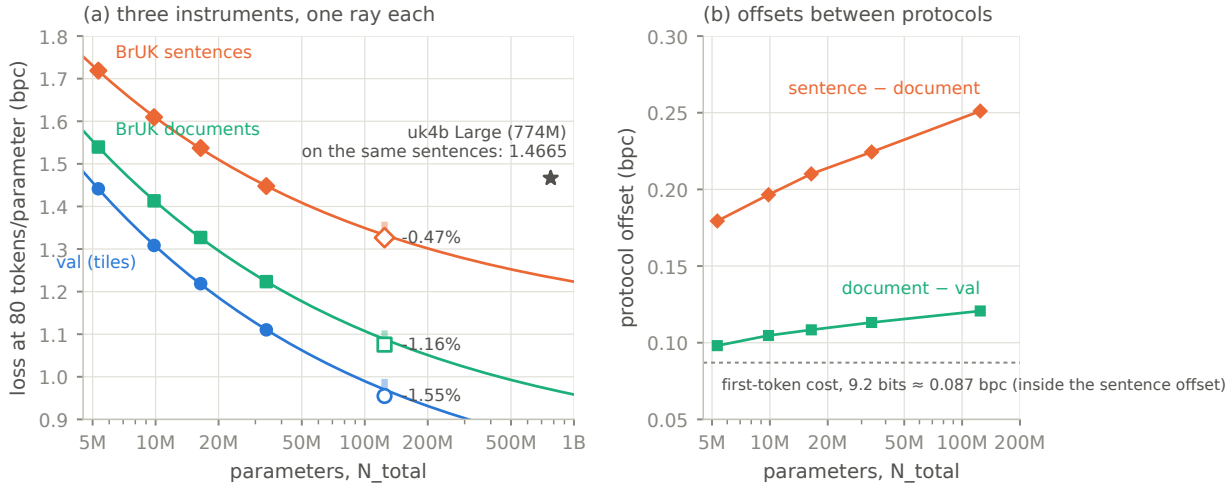


Figure 3. (a) Three instruments at 80 tokens per parameter — the full validation set, BrUK documents under the tile protocol, BrUK sentences under uk4b's protocol — each with a ray fit on the four rungs; the target's measured value (open marker) against the ray's prediction, whose leave-one-out range is the shaded bar. uk4b Large (774M) scored with its own code on the same 32,960 sentences: 1.4665. (b) The two protocol offsets grow with N; the first token of every sentence, predicted from BOS alone, costs ≈ 9.2 bits (0.087 bpc) at every size and sits inside the sentence offset.

uk4b. The previous from-scratch Ukrainian baseline reported 1.503 / 1.457 / 1.431 bpc for its 123M / 355M / 774M models on "BrUK29k", a 28,643-sentence subset filtered against their training corpus — the authors' corrected values

(repository commit 1c5dc381 and README erratum: the UNLP paper's figures were low by $1/(\ln 2)^2 \approx 2.08$, a nats-to-bits conversion applied in the wrong direction). Numbers on different subsets are not comparable, so we recovered their Large checkpoint and tokenizer from Internet Archive captures (2025-07-16 and 2025-08-23; the original host is offline), repaired the damaged leading bytes of both files (Appendix H), and scored the checkpoint with their `score.py` on our set: **1.4665**. Our 124M model at 9.9B tokens is 0.140 bpc lower than a model with 6.2× the parameters and ≈ 13× the training compute, our 124M model at 20 tokens per parameter (2.49B tokens, the own-rate run of §4.3) is 0.105 lower at 1.3617, and our 33.8M rung at 80 tokens per parameter is already 0.019 lower; interpolating the sentence ray, ≈ 29M parameters at that horizon would match it. At equal size the only number is their published Small (123M, 6.9B tokens) at 1.503 on their subset, 0.176 above ours, with the subset caveat. Three things favour them: their corpus (UberText 2.0: news, fiction, Wikipedia) is closer to BrUK's literary register than our web-dominated mixture; the corpus's native ASCII apostrophe is the form their tokenizer prefers (the typographic form costs them 0.045 on a 2,000-sentence sample); and any memorisation left in our set after filtering against *our* corpus favours them. Their published 1.431 on their own subset is 0.035 lower than on ours. This is a checkpoint comparison — corpus, tokenizer, architecture, optimizer and recipe all differ — and its dependence on the repaired tokenizer is discussed in Appendix H.

7 What re-measurement changed

The study began on 32 seed-locked windows and switched to the full 6,097-window set (190×) when a P3 anomaly failed to reproduce. The subset ran 0.027–0.031 bpc above full val at P1–P4 (two reconstructions; constant to ≈ 0.004 across 6.3× in N) — a hard draw, not a bug.

claim	instrument	full validation set	verdict
paired effect sizes (masking, QK-norm, residual scaling, …)	subset	moved ≤ 0.0005	survived
20× learning-rate vertices	subset	moved < 5%	survived
$\beta = 0.434 \cdot \ln(D/N) - 1.004$, $R^2 = 0.99$	subset	$R^2 = 0.064$	dead
learning-rate–duration exponents; width/duration decomposition	subset, 80× vertices	vertices outside their own grids	dead
separable L(N, D) surface (Chinchilla approach 3)	full val	$\chi^2 = 67$ on 7 df with independent equal-weight residuals (no covariance-aware version was computed; the verdict does not depend on it at this size); the N-term coefficient spans 15× across 6.3× in N	rejected
branch-endpoint $\sigma = 0.000074$	subset	0.00106	14× understated
"σ does not fall with scale"; three P3 anomalies	subset, n = 3	gone	artefacts
WSD beats cosine by 0.0793	untuned LR	cosine −0.0062 at tuned LR, one seed each	artefact
repetition penalty at four epochs	subset	within 0.004 of a continuation	withdrawn
batch asymmetry from Newton–Schulz noise promotion	—	present under AdamW	withdrawn
A falls with N and flattens at the target	full val	ladder 80× branches detuned (§4.4)	withdrawn

claim	instrument	full validation set	verdict
20× branch detuned by 0.005–0.009 bpc (model-based)	parabola in ln(lr) with ladder curvatures	own-rate run: 0.0057 lower	survived
sentence-protocol bpc 1.394 at the target, 0.073 below uk4b Large	packed scorer: first token unscored, mean ÷ cpt over all slots	re-scored under uk4b's exact rule: 1.327, 0.140 below	corrected
paragraph (line-level) protocol on a decontaminated set	the set scored was the unfiltered one (711 lines in the corpus)	—	dropped

Table 6. Claims by instrument.

The pattern is consistent and the mechanism specific. A constant offset leaves a fitted exponent unchanged, so constancy across rungs is not what killed the curvatures; the offset was not constant across horizons within a rung. On re-scoring, the single-seed P2 branches behind the retracted law moved their 20×→40× and 40×→80× gaps by 0.0016 each in opposite directions (0.0363 → 0.0347, 0.0218 → 0.0234), and β is a ratio of those gaps: at P2 it moves by 42, 106 and 64 per unit change in the 20×, 40× and 80× endpoints, so 0.0016 becomes 0.1–0.17 in β. Paired first-order differences compare two arms at the same horizon on the same windows, where the offset is common and cancels, which is why they moved ≤ 0.0005. A second instrument fault surfaced while preparing this revision: the packed scorer behind the sentence and line protocols dropped the first token of every unit and divided a per-scored-token mean by a characters-per-token ratio taken over all token slots, BOS and EOS included; re-scoring under uk4b's exact rule moved every sentence number down by 0.07–0.12 bpc, and the line-level set turned out to be the unfiltered one, so that protocol was dropped. Every curvature we now report carries its seed spread; quantities with unusually small variance are re-run first. Three had been flagged — Table 3's P2 interval and the masked P2 and P3 arms of §5.1 — and the per-seed values pulled from the run manifest resolve them: the masked 80× endpoints have seed sds of 0.0008 (P2) and 0.0006 (P3) over five seeds, and P2's per-seed β of 0.599 / 0.608 / 0.591 gives the 0.005 interval of Table 3; the smaller spreads quoted in earlier drafts were transcription errors. The checkpoint-level table of the two instruments' difference by rung, horizon and seed is pending (L.4).

A fine-tuning probe, and the same lesson. A small supervised fine-tuning of the 124M checkpoint (four Ukrainian task sets, 15,000 examples, loss on answer tokens only; Appendix M) first appeared to teach it nothing, and the conclusion "fine-tuning leads nowhere at this size" was assumed. But it was two instrument faults. Under document masking a BOS token the model opens a new segment and masks attention to the prompt, so sampled answers were written blind to their questions; and the grammatical-error-correction set, whose targets are ≈ 90% verbatim copies of their inputs, held 49% of the supervised tokens, so about 44% of the gradient was teaching the model to repeat its input. Forbidding BOS at sampling and down-weighting copied text (a token-level diff of source and target) produced the expected profile for 124M parameters: answer formats and register acquired, facts mostly absent, looping present. Two further observations shape Part 2's capability curve: the first answer token carries most of uncertanty and up-weighting it made held-out loss monotonically worse — the gradient already concentrates at early positions (Qi et al. 2024) — so per-task first-token accuracy is reported alongside held-out loss.

8 Preparing 500M and 1B

Record. The four-rung ray registered before the 124M run gives 500M → 0.900 and 1B → 0.860 at 20×, masked, with leave-one-out ranges [0.882, 0.932] and [0.836, 0.899]; its 124M prediction was met at −0.3% by the branch and missed by 0.0017 below the leave-one-out range by the run at its own rate (§4.3). **Forecast for Part 2.** The refit registered with the previous revision added the 124M 20× branch point: 500M → 0.894, 1B → 0.853 (leave-one-out [0.892, 0.902] and [0.849, 0.862]); it stays in the record. The like-for-like refit — the ladder's 20× points and the target's 20× point all at their own vertices — adds 1.0005 instead: **500M → 0.883, 1B → 0.839** (leave-one-out [0.878, 0.902] and [0.831, 0.862]; 512M → 0.881), and is the forecast Part 2 is held to. The four-rung values stay in the record; forecasts will be restated at the final parameter counts. The 20×→80× gain at 124M is 0.0456 as measured (own-rate 20× to 80×); extrapolating it to other horizons needs the A, β split of Table 3, 0.081–0.089 and 0.52–0.60 across the 40× correction, and the ladder's own horizon data are not used for it (§4.4).

9 Limitations

One language, one corpus, one tokenizer, one context length, sizes up to 124M. The configuration comparisons of Appendix C were made at the pre-freeze P1 configuration on the earlier instrument. Loss-only, apart from a qualitative fine-tuning probe of the 124M checkpoint (Appendix M); the capability curve across rungs is Part 2. The target is a single seed; its 20× point is measured at its own rate (one run) and its branch machinery verified from scratch at one seed, its 40× point is a detuned branch with a model-based correction, so its β is a range. The ladder's own 40× and 80× branches are detuned by rung-dependent amounts (measured at P2 and P3 only), so per-rung horizon fits are reported as measured and no trend in A or β is claimed. The repetition result is a consistency check. The batch-size accounting is at P1 only; the per-knob decomposition is under AdamW, and the Muon+AdamW recovery is one joint arm (two rate variants) against a three-seed reference, with intervals of ± 10 points. Two estimates of the masking effect differ by up to 0.001 at P2 and by 0.0016 at P1, where they were taken at different learning rates. The Muon wall-clock verdict is implementation-specific. The uk4b comparison depends on a repaired tokenizer whose validation is described but whose token-id fixtures are not yet released, and is on our sentence set filtered against our corpus. Sentence-level protocols understate larger models on this ladder, including ours.

10. AI assistance and author contributions

This study was directed and executed by the author, who designed pipelines and infrastructure, implemented models and ran the experiment. An AI assistants (Claude, Anthropic + ChatGPT, OpenAI + Gemini, Google) used for drafting specialized scripts, used for verification and debugging pipline scripts, and writing UIs and custom features (report collection and ran statistic computation scripts). An AI + Google search (with Gemini) was used to lookup for references and pinpoint previous finding at specific question during the study, combination of different family model was used to review and revise this paper. The author is responsible for all claims and errors. Interaction records were retained.

Data and licence statement

FineWeb-2 is released under ODC-By 1.0 and its use is subject to Common Crawl's terms; Ukrainian Wikipedia under CC BY-SA 4.0; both are attributed here. The Brown-Ukrainian corpus (CC BY-NC-SA 4.0) was not used as training data; 7.35% of its sentences occur verbatim in the FineWeb-2 portion of our corpus and were excluded from the sentence protocol. Our BrUK sentence set is released as SHA-1 hashes with the construction script rather than as text. The fine-tuning sets of Appendix M are used under their own licences (ua_datasets MIT; UA-GEC CC BY 4.0; Aya Dataset Apache-2.0) and are not redistributed. The uk4b checkpoint and tokenizer were recovered from Internet Archive captures of the authors' public release and used only to score the authors' model.

Reproducibility statement

Released under Apache-2.0: training and evaluation code, all ladder and target checkpoints including the cooldown branches, the fit scripts with their inputs, the uk4b scoring patch and the byte-level repair scripts. Also released: the validation set as document ids and hashes with the tile protocol; the BrUK sentence hashes, filters and flags; per-seed endpoint tables (Appendix D); the run manifest; and the pre-registration record (Appendix K). Items still to be generated are listed in Appendix L.

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Appendices

A. Cleaning pipeline

The normaliser: strip non-orthographic stress marks via NFD → mark removal on Cyrillic bases → NFC (naive orderings destroy й and ї or leave non-NFC output); fold six apostrophe code points (U+2019, U+02BC, U+2018, U+0060, U+00B4, U+02B9) to U+0027, since the apostrophe is orthographic in Ukrainian; remove invisible characters (ZWJ/ZWNJ, soft hyphen, BOM, bidi controls) before whitespace collapse; normalise whitespace while keeping paragraph breaks; scrub Wikipedia image-markup residue anchored on patterns never legitimate in prose. Ordering: marks, then whitespace, then markup. Five heavier rules — homoglyph repair on glued words, a case-glue quality signal, namespace-lead detection, same-script CamelCase splitting, dropping bilingual documents — were built and removed because their false positives (ordinary orthography such as кВт, КЗпП, ПриватБанк; Ukrainian inflection) outnumbered real artefacts about 10:1. On the curated Brown-Ukrainian corpus the cleaner changes 73 of 36,038 sentences and −29 of 3.64M characters.

B. Tokenizer validation gates

Five gates every candidate vocabulary had to pass: ids fit in uint16; digit ids identical across six surrounding contexts; 2,000 held-out documents round-trip byte-exact; newline-bearing pieces present (30 / 48 / 76 / 107 at 8k / 16k / 32k / 49k); BOS and EOS bracket every document. Full-corpus characters per token matched the held-out sweep to within 0.6%.

C. How the configuration was chosen

component	chosen	alternative tested	Δ bpc (alternative − chosen; positive favours the choice)	seeds	instrument
residual scaling	off	GPT-2 1/√(2L) on	+0.0099 (P1); +0.0048 [0.0041, 0.0056] (P2)	factorial, 1; 5 paired	subset (C.2)
biases	none	with biases	+0.0003 (below MDE)	factorial, 1	subset (C.2)
initialisation	0.55/√fan_in	PyTorch default (kaiming); fan-in-corrected variant	−0.0029 (below MDE); +0.0012 [−0.0004, +0.0029]	factorial, 1; 8 paired	subset (C.2)
QK-normalisation	on	off	+0.0125 (P1, p = 0.011); +0.0118 [0.0065, 0.0170] (P2)	3 paired; 3 paired	subset (C.3)
MLP	SwiGLU, h = 320 (≈ 8d/3)	GELU, h = 480 (3.75d; equal parameters)	−0.0019 [−0.0054, +0.0017]	5 paired	subset (C.4)
normalisation	RMSNorm	LayerNorm	+0.0028 [−0.0008, +0.0063] (5/5 same sign)	5 paired	subset (C.4)
optimizer	Muon + AdamW	AdamW everywhere, own vertex	+0.0096 (P1); +0.0077 (P2); +0.0170 (frozen, full val, both masked)	5 vs vertex; 5 paired; 5 vs 3	subset; subset; full val (C.5)
schedule	WSD, 20% linear	cosine at own vertex	−0.0062	1	subset (C.6)
batch	32,768 tok/step	16,384; 65,536	see §5.2	5 paired; 3 paired	full val (C.7)
document masking	on	off	+0.003 … +0.006 (P1–P4; two estimates, §5.1)	paired	full val
RoPE reset	off	on	≈ 0 (algebraically inert under the mask)	paired	full val
context	2,048	512; 1,024 (eval truncation); 4,096 (extrapolated)	+0.0372; +0.0139; ≈ −0.008 vs +15% FLOPs	1 (P2 ckpt)	subset (C.8)
vocabulary	32k	8k / 16k / 49k	fertility sweep, §2	—	held-out (C.9)
weight decay	0.1 on 2D	—	inherited, not swept; C.10	—	—

Table C.1. Configuration decisions. Sign convention alternative − chosen throughout; a negative entry means the alternative was nominally better and the choice was kept on other grounds. P1 comparisons are against the eight-seed P1 baseline at the same seeds; P2 against the five-seed P2 baseline. The biases and initialisation rows carried the opposite sign in draft v3; the values here are read from the run-level factorial (C.2).

All decisions were taken at P1 (5.3M parameters, 20 tokens per parameter) and re-checked at P2 before the configuration was frozen, on the 32-window instrument unless stated. The selection heuristic was a paired seed difference exceeding the design sensitivity at the seeds used, each arm at its own learning-rate vertex; anything below that was kept on principle (fewer parameters, simpler code, convention, or a scale-aware property) and is labelled as such. The P1 baseline for every comparison is eight seeds (47–54) at the tuned rate: 1.5562 bpc, σ = 0.00237, 95% CI [1.5543, 1.5582]; paired comparisons use the five seeds 47–51 (mean 1.5556); the P2 baseline is five seeds (47–51), mean 1.3998.

The overall accounting at P1: the first 32k-vocabulary run (cosine, lr 1e-3) scored 1.7936 bpc; tuning the learning rate and switching to WSD took it to 1.5562; dropping residual scaling and adding Muon took it to 1.5374 (five seeds, 95% CI [1.5321, 1.5427]). Of the 0.256 bpc total, 93% came from the learning rate and schedule and 7% from architecture and optimizer combined. The additivity of the two effects (residual scaling off; Muon) was tested at five seeds: predicted −0.0194 from the components, measured −0.0182 against the paired five-seed baseline (1.5556).

C.1 Learning rate. With two exceptions noted below, no comparison in this paper was made at a shared learning rate. Each arm was swept on a log-spaced grid (×2 spacing, three to five points, widened whenever the best point sat at an edge); a parabola was fit in ln(lr) on the points within a ~4× span around the minimum and the vertex was used. Three facts from the sweeps shaped everything else. (i) The seed standard deviation at the vertex is 0.00237 bpc against 0.016 at one-twelfth of the vertex: a mistuned learning rate inflates seed noise roughly sevenfold, and produced the retracted schedule result (C.6). (ii) The vertex moves with configuration — removing residual scaling halved it (C.2) — so a flag change without a re-sweep measures the wrong thing. (iii) A grid wider than ~4× breaks the parabola: at P1 the 1.5e-3 point sat 10σ above the minimum and dragged the five-point vertex to 6.7e-3 (residual 0.0036); dropping it gave 5.10e-3 with residual 0.00003. Three P1 AdamW rates appear in this paper and are different objects: 4.93e-3 is the embedding-group rate of the Muon+AdamW configuration (Table 1); 4.8e-3 is the AdamW-everywhere arm's vertex at the pre-freeze configuration on the subset (C.5); 5.10e-3 is the AdamW-everywhere vertex at the frozen configuration on the full validation set, the reference rate in §5.2. The two shared-rate comparisons are the retracted schedule comparison (C.6) and the P2 optimizer comparison at the grid rate (C.5), which is also given at the fitted vertex.

C.2 Residual scaling, biases, initialisation (a 2^3 factorial). The three GPT-2-inherited choices were separated with a full factorial at P1: 8 cells × 3 learning rates (2e-3 / 4e-3 / 8e-3; 24 single-seed runs, seed 47), each cell read at its own best rate. Main effects, each pooling four cells (SE 0.0017, threshold 0.0040), as on − off for each flag: residual scaling on +0.0099 (1.5486 → 1.5585), the only effect above threshold; the custom initialisation $0.55/\sqrt{\text{fan_in}}$ on +0.0029 (1.5521 → 1.5550), i.e. the PyTorch default was nominally better but below threshold, as predicted from the two being within 5% of each other at fan_in = d; biases off −0.0003 (1.5537 → 1.5534), nothing. Removing residual scaling moved the optimal learning rate down by 2× (all four residual-off cells preferred 4e-3, three of four residual-on cells 8e-3), consistent with the $1/\sqrt{(2L)}$ output-projection shrink being a 3.46× factor at L = 6 where GPT-2 calibrated it at L = 12–48. The best single cell was no biases, default initialisation, no residual scaling (1.5464 at 4e-3), 0.0039 below the chosen cell at one seed. The residual-scaling effect was re-tested at P2 with five paired seeds: +0.0048 bpc [95% CI 0.0041, 0.0056], t = 17.8, half the single-seed P1 main effect but far from zero, with training loss also worse (+0.0041, p = 0.03), i.e. an optimisation cost at this depth rather than regularisation; whether the ranking holds at longer horizons was not tested. Because $1/\sqrt{(2L)}$ shrinks harder with depth (0.19 at L = 14), the effect was scheduled for a P4 re-test that was not run. A fan-in-corrected initialisation against the flat-σ $0.55/\sqrt{\text{fan_in}}$ convention (which deviates only at the MLP output projection, by $\sqrt{(h/d)}$) gave +0.0012 [−0.0004, +0.0029] over eight paired seeds. Decisions: residual scaling off on the evidence; no biases kept for fewer parameters; $0.55/\sqrt{\text{fan_in}}$ kept, despite the default's nominal −0.0029, because it fixes the initial logit scale across widths, which the sweep design relies on.

C.3 QK-normalisation. Off costs +0.0125 at P1 (three paired seeds: +0.0103 / +0.0149 / +0.0124; t = 9.4, p = 0.011) and +0.0118 at P2 (three paired seeds, 95% CI [0.0065, 0.0170], t = 9.7) — the same size at both rungs, as expected for a mechanism tied to the constant head dimension of 64. The ablated arm has 2.1× the seed variance at P1, and at P2 its training loss degrades twice as much as its validation loss (+0.024 vs +0.012), both signatures of an optimisation aid rather than a representational change. Its cost is 768 parameters at P1 and 1,024 at P2 ($2 \cdot L \cdot 64$), and 1.9% wall-clock at P2. Kept.

C.4 MLP and normalisation. GELU with h = 480 = 3.75d (two matrices) against SwiGLU with h = 320 (three matrices) at equal parameters ($2 \cdot 128 \cdot 480 = 3 \cdot 128 \cdot 320 = 122{,}880$), five paired seeds at each arm's vertex: GELU − SwiGLU = −0.0019 [−0.0054, +0.0017], p = 0.21, three of five seeds negative; the interval excludes effects beyond 0.0054 in GELU's favour and 0.0017 in SwiGLU's. LayerNorm against RMSNorm, five paired seeds: LayerNorm − RMSNorm = +0.0028 [−0.0008, +0.0063], p = 0.097, all five seeds positive (exact sign test, two-sided p = 0.06) but one seed carrying 56% of the total; directionally in RMSNorm's favour, below the design sensitivity. Both choices kept — SwiGLU for convention, RMSNorm for cost and because it was directionally favoured. Under the flat-σ initialisation the MLP output projection's effective scale differs between the two MLP arms by $\sqrt{(h/d)}$ (1.58× vs 1.94×), so the comparison is between two recipes rather than between activations alone; the fan-in-corrected initialisation removes the difference and was itself null (C.2).

C.5 Optimizer. Muon on the hidden 2D matrices with AdamW on the embedding and 1D parameters, against AdamW on everything, each at its own vertex. At P1 the Muon arm's sweep (0.002 / 0.006 / 0.01 / 0.03) has its vertex at 0.0082–0.0087 and is flat between 0.006 and 0.01; the AdamW-only arm's five-point sweep has its vertex at 4.8e-3. Muon wins by 0.0096 at P1 (five-seed Muon mean 1.5374 vs the AdamW-only parabola at its vertex, 1.5470) and by 0.0077 at P2 (five paired seeds at the grid rate 4e-3 give +0.0090 with sd 0.0023; at the fitted vertex 0.0077). An earlier figure of −0.0119 at P1 compared Muon against AdamW at 8e-3, 1.7× that arm's vertex, and is superseded. At the frozen configuration on the full validation set the gap is −0.0170 (1.5001 over five seeds vs 1.5171 over three; the AdamW arm ran with every other setting frozen, including the mask, confirmed from the run configuration; had it run unmasked the comparison would have been against 1.5048 and the gap −0.0122). Two further facts: Muon's optimal learning rate did not move when residual scaling was removed (vertex 0.0082 → 0.0087), because orthogonalisation fixes the update magnitude independently of weight scale, whereas AdamW's vertex halved; and the Muon configuration's seed sd at five seeds is 0.0042 against the baseline's 0.0024 (1.8×; 3.2× in variance). Muon's wall-clock overhead at P1 is 19.9% in this implementation, dominated by kernel dispatch of the Newton–Schulz iterations, and about 1% at the target.

C.6 Schedule. WSD against cosine was first compared at a shared learning rate of 1e-3, about one-twelfth of either schedule's vertex: WSD won by 0.0793 bpc over eight seeds (t = 25). At each schedule's own vertex — cosine 1.5503 at 1.6e-2, WSD 1.5565 at 1e-2, one seed each — cosine is nominally better by 0.0062, a single-seed comparison that neither establishes nor excludes parity; the untuned result is listed among the retractions (§7). WSD was retained because cooldown branching yields the loss at several horizons from one trunk, validated at P1 (seeds 47–49: a 20× branch reproduced the from-scratch run to +0.0000 / +0.0002 / −0.0002); at P2–P4 only mean-level agreement between three-seed from-scratch runs and five-seed branch endpoints is available (§4.1), and per-seed gates there are listed in Appendix L.4. The decay fraction (20%; 18.5% in the earliest P1 runs), the floor (0.1× peak) and the warmup (200 steps) were inherited rather than swept; §5.2 finds no detectable effect of warmup length at P1 (+0.0011 [−0.0085, +0.0107]).

C.7 Batch size. 32,768 tokens per step was chosen from a P1 sweep over 16,384 / 32,768 / 65,536 at fixed tokens with step-defined hyperparameters (five paired seeds, unmasked configuration). §5.2 shows that sweep measured the schedule: with warmup, weight-decay timescale and EMA horizons held constant in tokens, 16,384 is 0.0028 better at fixed tokens under AdamW. The frozen value stands for wall-clock reasons; its definition has been changed to token-based horizons for Part 2.

C.8 Context length and sequence composition. T = 2,048 was set by a cost rule checked against two measurements. The rule: at the target, T = 4,096 costs ≈ +15% FLOPs per token (attention scores grow with T against the matrices and the output projection; +18% without the output projection), so at the 20× horizon it had to buy ≈ 0.007–0.008 bpc to break even; at the 80× horizon the target trained to, the same 15% of tokens costs ≈ 0.003. The first measurement is the P2 checkpoint evaluated at truncated windows: T = 512 costs +0.0372, T = 1,024 +0.0139, and extrapolating the per-position curve to 4,096 gives ≈ −0.008 — a wash at 20× and a small gain at 80×. The second is the position-bucketed loss across three rungs (per-window accumulation over all 6,097 windows, log-spaced buckets): the tail slope from bucket [512–1023] to [1024–2047] is −0.0047 at P1, −0.0044 at P2 and unchanged at P3, and the P1→P2 advantage peaks at positions 64–128 and is flat beyond — scale buys short- and mid-range context and nothing at the tail at these sizes. Part of that flatness is data-side: under contiguous packing, own-document context grows only 1.52× when window position doubles from 512 to 1,024, and at position 2,047 about 57% of the attended context belongs to a different document (Zhao et al. 2024; Ding et al. 2024). That arithmetic motivated block-diagonal masking, which recovers the cross-document waste at no FLOP cost and was measured directly (§5.1). The decision therefore rests on the tail data and on masking rather than on the break-even arithmetic alone; it was reopened once, when a since-retracted β law predicted the horizon term would favour 4,096 at the target, and closed again on the three-rung tail data. The sentence-minus-document offset of §6 is a protocol difference and is not used as evidence here. The choice is revisited for ≥ 500M, where Kobza's document lengths are larger.

C.9 Tokenizer. The vocabulary sweep of §2 and the gates of Appendix B. 32k was chosen because the 16k→32k doubling still gained 10.8% in characters per token while the next (extrapolated 9.2%) did not clear 10%, and because a 49k embedding would be 85% of P1's parameters (32k is 79%) with 1.5× the logit memory per step.

C.10 Weight decay. 0.1 on all 2D tensors including the tied embedding, 0 on 1D parameters, is the nanoGPT convention (minGPT excluded the embedding). It was not swept at P1; the accounting in §5.2 shows its timescale to be one

of the step-defined knobs that produced the batch artefact, and a sweep at the token-defined setting is queued.

C.11 Depth and shape. 14 layers rather than 12 follow from a 100M matrix-parameter design target (§3); the ladder's rung shapes keep head dimension 64 and h ≈ 8d/3 rounded to a multiple of 64 so that per-layer matrix parameters are $4d^2 + 3dh$ at every rung.

C.12 Instrument, seeds and pairing. The 32-window subset ran above the full validation set by 0.027–0.031 bpc at P1–P4 (0.0301 / 0.0305 / 0.0269 at P1–P3 from the per-window passes; 0.0296 / 0.0293 / 0.0288 / 0.0269 at P1–P4 from the masking arms), a rung-constant offset that preserved paired differences and, because it was not constant across horizons, destroyed ratios (§7). Same-seed reruns of an identical configuration differ by 0.0005–0.0010 bpc, so seed variance dominates nondeterminism at P1 and is comparable to it at P2–P4. Seed pairing survives an initialisation change (residual arm: sd of the paired difference 0.0006) but not a change to the computation graph (QK-norm and optimizer arms: 0.0021–0.0023); power calculations used the pessimistic figure whenever an arm altered what the model computes. Single-seed picks inflated three times during the study (the best P1 seed sat 1.2σ below the five-seed mean; a single-seed SwiGLU effect went from −0.0089 to −0.0019 across three measurements), which is why every number in the body is a seed mean where seeds exist; the exceptions are labelled where they occur.

D. Per-seed tables

D.1 Baselines. Causal 20× from scratch, P1 seeds 47–51: 1.5007, 1.5074, 1.5031, 1.5059, 1.5071. Masked 80× branch endpoints per seed (full validation set; seeds 47–51 at P1–P3, 47–49 at P4): P1 1.4394, 1.4416, 1.4404, 1.4425, 1.4430 (mean 1.4414, sd 0.0015); P2 1.3084, 1.3071, 1.3088, 1.3084, 1.3092 (1.3084, 0.0008); P3 1.2178, 1.2190, 1.2187, 1.2185, 1.2193 (1.2187, 0.0006); P4 1.1103, 1.1097, 1.1106 (1.1102, 0.0005) — the means of Tables 3 and 5. Masked P2 20× and 40× per seed (47–49): 1.3665, 1.3649, 1.3662 and 1.3315, 1.3300, 1.3317; P2's per-seed three-point β: 0.599, 0.608, 0.591 (SE 0.005). P2–P4 causal per seed and the remaining masked 20×/40× endpoints: [from the run manifest; means in §4.1].

D.2 Document-masking arms (full val, 20×, paired by seed).

rung	causal (seeds 47, 48, 49)	masked (47, 48, 49)	paired Δ	configuration
P1	1.5175, 1.5201, 1.5181	1.5106, 1.5135, 1.5127	−0.0069, −0.0066, −0.0054 (mean −0.0063)	pre-re-tune learning rate
P2	1.3694, 1.3713, 1.3719	1.3664, 1.3664, 1.3663	−0.0030, −0.0049, −0.0056 (−0.0045)	frozen; masked spread under verification (L.10)
P3	1.2741, 1.2757, 1.2760	1.2718, 1.2719, 1.2721	−0.0023, −0.0038, −0.0039 (−0.0033)	frozen; masked spread under verification (L.10)
P4	1.1630	1.1600	−0.0030	frozen, seed 47

RoPE-reset arms (masked + reset): P1 1.5106, 1.5132, 1.5126; P2 1.3660 (seed 47); P3 1.2725 (47); P4 1.1601 (47). A second evaluation pass of the P1 masked arm extended to five seeds lists 1.5106, 1.5133, 1.5130, 1.5139, 1.5125; seeds 48 and 49 differ from the first pass by 0.0002–0.0003 (a rerun, so the difference is run-to-run nondeterminism). The five-seed masked and causal sets behind estimate (b) of §5.1 are the branch endpoints and from-scratch runs of §4.1.

D.3 Batch arms at P1 (full val, fixed 106.5M tokens; run manifest `p1_adamw_batch`, `p1_re_adamw_arm`).

Muon+AdamW, unmasked configuration, seeds 47–51: mbs 8 → 1.5295, 1.5386, 1.5309, 1.5299, 1.5328 (mean 1.5323); mbs 16 → 1.5007, 1.5074, 1.5031, 1.5059, 1.5071 (1.5048); paired halving +0.0288, +0.0312, +0.0278, +0.0240, +0.0257 (mean +0.0275, sd 0.0028, t = 22.1); mbs 32 → 1.5022, 1.5090, 1.5059, 1.5067, 1.5097 (1.5067); paired doubling +0.0015, +0.0016, +0.0028, +0.0008, +0.0026 (mean +0.0019, t = 5.0).

AdamW everywhere, frozen configuration, seeds 47–49: mbs 8 → 1.5400, 1.5417, 1.5441; mbs 16 → 1.5160, 1.5183, 1.5169; mbs 32 → 1.5153, 1.5204, 1.5181. Single-knob arms at mbs 8: wd 0.05 (lr 5.10e-3) → 1.5344, 1.5344,

1.5320; warmup 400 → 1.5420, 1.5419, 1.5387; lr 3.61e-3 (wd 0.1) → 1.5372, 1.5401, 1.5420; β (0.95, 0.975) → 1.5291, 1.5253, 1.5349; all four (lr 3.61e-3, wd 0.0707, warmup 400, β .95/.975, as recorded in the run table) → 1.5127, 1.5159, 1.5141.

Muon+AdamW joint arm at mbs 8, frozen configuration (AdamW 3.50e-3, wd 0.0707, warmup 400, Adam β .95/.975, Muon momentum 0.975), seeds 47–49: Muon 5.79e-3 (√B) → 1.4989, 1.5040, 1.5000 (1.5010); Muon 8.16e-3 (unscaled) → 1.5007, 1.5032, 1.5031 (1.5023). Masked mbs-16 reference (`p1_basesweep2`, AdamW 5.10e-3, Muon 8.16e-3), seeds 47–51: 1.4979, 1.5011, 1.4996, 1.5015, 1.5001 (mean 1.5000, sd 0.0014); masked mbs-32, same settings: 1.4991, 1.5031, 1.5005, 1.5012, 1.5001 (1.5008; paired doubling +0.0008, t = 1.8). Masked mbs-8 Muon+AdamW reference at the frozen configuration (run `paper1_final_r1`, 2026-10-05, seeds 47–49, AdamW 4.93e-3): 1.5282, 1.5301, 1.5300 (1.5294, sd 0.0011); paired halving penalties 0.0303, 0.0290, 0.0304 (mean 0.0299); paired recoveries √B 0.0293, 0.0261, 0.0300 (mean 0.0285), unscaled 0.0275, 0.0269, 0.0269 (0.0271).

D.4 BrUK per seed (80× endpoints; seeds 47–51 at P1–P3, 47–49 at P4). Sentences (uk4b rule, re-scored 2026-10-05): P1 1.7139, 1.7219, 1.7165, 1.7214, 1.7209; P2 1.6057, 1.6090, 1.6107, 1.6113, 1.6112; P3 1.5372, 1.5361, 1.5342, 1.5397, 1.5390; P4 1.4483, 1.4474, 1.4478; target 80× 1.3267, target 20× at its own rate 1.3617, target 20× from scratch at 0.45e-3 1.3667. Documents, target 20× at its own rate: 1.1186. The earlier draft's sentence and line-level values, scored without the first token and with the mixed denominator (§7), are retained in the released score files, not here. Documents: P1 1.5369, 1.5396, 1.5396, 1.5420, 1.5393; P2 1.4140, 1.4120, 1.4118, 1.4132, 1.4145; P3 1.3264, 1.3270, 1.3266, 1.3272, 1.3281; P4 1.2229, 1.2230, 1.2242.

D.5–D.6 Instrument difference by checkpoint and the seed-bootstrap band for the ray: pending (L.4).

E. Learning-rate curvatures and Test A

Fitted parabola curvatures a (bpc per ln² of Muon learning rate), from the single-seed sweeps on the 32-window instrument: P2 0.0120 at 20× (vertex 5.54e-3) and 0.0192 at 80× (vertex 4.20e-3; three points); P3 0.0137 at 20× (vertex 3.66e-3) and 0.0127 at 80× (vertex 1.94e-3; four points, residual 1e-4). The registered correction of §4.3 used a = 0.014–0.018, a range inside the span 0.0120–0.0192 of these fits; §4.3 quotes the full span and an extrapolation. The 20× vertices re-measured on the full validation set moved < 5% (P1 AdamW 5.10e-3 → 4.93e-3, P2 Muon 5.54e-3 → 5.45e-3, P3 Muon 3.66e-3 → 3.63e-3); the curvatures quoted here are the subset-instrument fits and were not re-fit — the full-validation re-scoring re-fit the vertices only. The runs used the sweep points of Table 1 (P2's trunk 5.00e-3, 8% below its full-validation vertex, the others within 4%), which is why §4.4 prices the P2 branch detuning at the trunk's rate. Horizon exponents $\gamma = \ln(\mathrm{lr}_{80}/\mathrm{lr}_{20})/\ln 4$: −0.20 at P2, −0.46 at P3. Test A: P2, seed 47, horizons 20 / 28.3 / 40 / 56.6 / 80× (ratio √2), subset instrument, L = E + A·r^−β with three parameters: β = 0.620, residuals +0.0001 / −0.0003 / +0.0002 / +0.0001 / −0.0001; the three-point fit on 20 / 40 / 80× predicted the two held-out horizons to −0.0005 and +0.0001.

F. Branching gates and the junk test

Branching. A branch copies the trunk at 80% of a horizon and runs the 20% linear decay to 0.1× peak; at P1 (seeds 47–49) a 20× branch reproduced the corresponding from-scratch run to +0.0000 / +0.0002 / −0.0002; at the target (seed 47, 20×, Muon 0.45e-3) the branch scored 1.0062 against 1.0061 from scratch (BrUK sentences 1.3667 from scratch); at P2–P4 the mean-level comparison of §4.1 is available and per-seed gates are pending (L.4).

Junk test. The causal-trained P4 (seed 47) evaluated under the document mask minus under causal attention, bucketed by the number of foreign (previous-document) tokens preceding each scored token in its tile: the tile's leading truncated segment 0 by construction; 0–1 foreign tokens ≈ 0; 2–1,023 foreign tokens a plateau of 0.014–0.020 nats per token (0.005–0.007 bpc); the [1024–2047] bucket 0.032 nats (0.012 bpc); overall +0.0138 nats per token = +0.0049 bpc. The cost does not grow with the amount of foreign context between 2 and 1,023 tokens, which is consistent with a per-document mismatch (a BOS attending only to itself never occurs in causal training) rather than lost content; a benefit that saturates within a few tokens cannot be excluded by this one-sided test, and the buckets differ in other respects. The last bucket is unexplained.

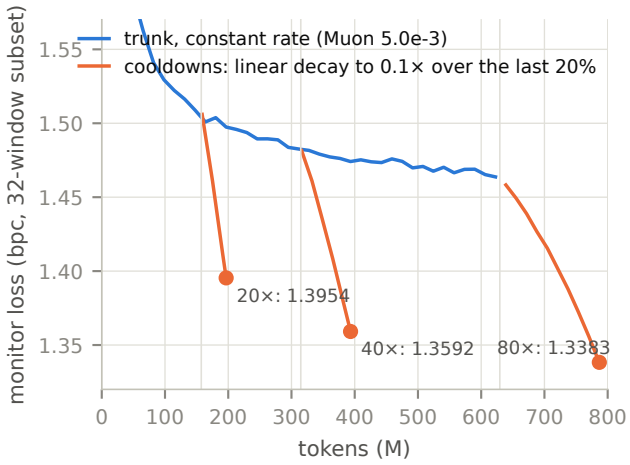


Figure F1. P2, seed 47: the trunk at its constant rate (Muon 5.0e-3) and the three cooldowns, each a copy of the trunk at 80% of its horizon decayed linearly to 0.1× over the remaining 20%. The curve is the during-training monitor (the retired 32-window subset, ≈ 0.03 above the full set); the endpoints on the full validation set are those of Table 3 (1.3665 / 1.3315 / 1.3084 for this seed).

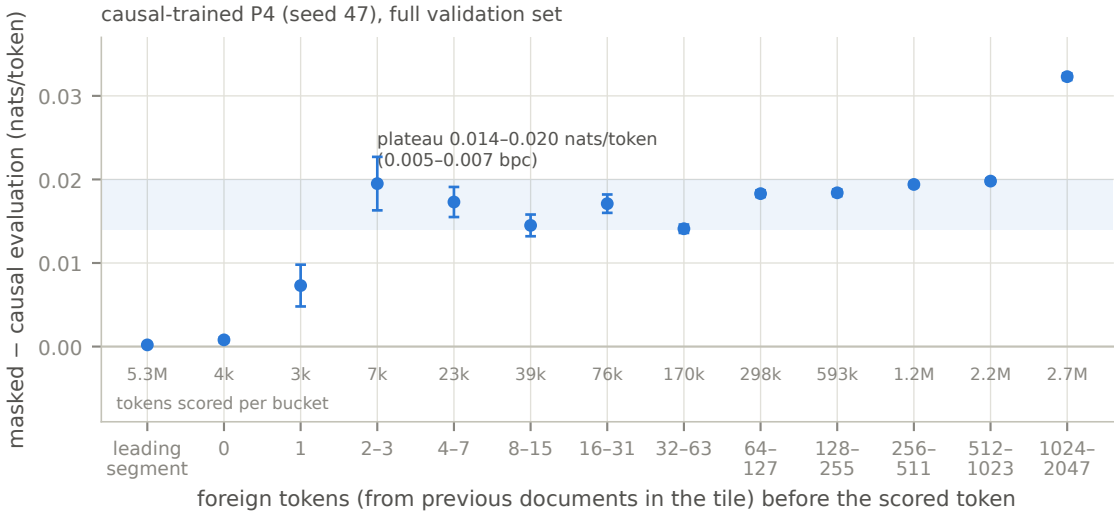


Figure F2. The junk test. The causal-trained P4 (seed 47) scored under the document mask minus under causal attention, per token, bucketed by the number of foreign tokens from previous documents that precede the scored token within its tile (bars: standard errors; token counts per bucket below). The leading truncated segment is 0 by construction; a plateau of 0.014–0.020 nats per token (0.005–0.007 bpc) from 2 to 1,023 foreign tokens; the [1024–2047] bucket at 0.032 is unexplained.

G. BrUK dataset construction

Sentences: 36,038 from `tokenize_uk` → 463 SHA-1 duplicates → 35,575 unique → 2,615 exact-containment hits against the training corpus via Aho–Corasick on identically cleaned text (7.35%: 1,421 ≤ 5 words, 248 of 6–8, 946 ≥ 9) → 32,960 kept; a loose match (lower-cased, punctuation stripped) finds 3,832 (1,217 beyond the exact hits) and is reported only; MinHash (word 3-grams, 32 permutations, Jaccard ≥ 0.8, ≥ 8 words) 60 hits, 6 new (1,680 characters, 0.048%), not dropped. Lines (protocol dropped, §7): 11,633 native lines, 11,582 unique, of which 711 (6.1%) occur verbatim in the training corpus and 3 more by MinHash (40 of 9,968 checked); the set scored in earlier drafts was the unfiltered 11,633. The document protocol applies no sentence-level filter. Full-document hashes 0 of 805. Scores under the sentence protocol are pooled bits over pooled characters (micro-average), not a mean of per-sentence ratios; the first token of each sentence is scored from BOS and the end token is not, as in uk4b's `score.py`; the per-model first-token sums are released with the scores. Cleaner on BrUK: 73 of 36,038 sentences changed, −29 characters; 76 of 11,633 lines; 13 curly apostrophes, 11 acute accents, tabs and NBSPs. Released as hashes, flags and the script.

H. uk4b recovery, repair and scoring

Artefacts. The final published checkpoint (`ckpt10m.pt`, 36 layers, d 1,280, 20 heads, block 1,024, vocabulary 50,257; `iter_num` 10,465,280 ≈ 21.4B tokens) and the SentencePiece model (`wiki.model`) were recovered from Internet Archive captures (2025-07-16 and 2025-08-23); the origin host returns 404 and the Small and Medium checkpoints were never captured. Both files arrived with their first bytes overwritten by an HTTP error page after a resumed download.

Checkpoint. The 146-byte prefix was rebuilt from the zip central directory, the first entry's data descriptor and the pickle preamble, and verified by CRC: the rebuilt prefix plus the surviving bytes reproduce the stored CRC of `ckpt-2/data.pkl`, and every entry of the 9,439,979,901-byte archive passes `testzip`. The weights themselves were never damaged.

Tokenizer. The lost 143 bytes held the first six vocabulary entries — `<pad>` (id 0) and five control pieces (ids 1–5) — and the first three bytes of the seventh (`<|endofmeta|>`, id 6), whose text, score and type survived. Everything from id 6 onward, including every text piece, its score and type, and the model's `normalizer_spec`, is intact bytes. The repair is determined, not guessed: (1) the number of lost entries is fixed by `trainer_spec.vocab_size` (50,257) minus the surviving count, giving six; (2) their identity and order are fixed by `trainer_spec` (pad id 0; control symbols take the lowest free ids in listed order), which places `<|endofmeta|>` at index 6, matching the surviving partial entry; (3) the six rebuilt entries serialise to exactly 143 bytes, the length of the lost region; (4) the repaired model loads with 50,257 pieces, `</s>` at 50,256 (unk/bos/eos 50254–50256 per `trainer_spec`), `<pad>` at 0 and `<|endofmeta|>` at 6, and round-trips Ukrainian text. Control pieces are never produced by segmentation of raw text and their scores (set to 0) do not enter it, so text segmentation and every text piece's id are those of the original model; the remaining eleven control ids (1–11) are the five rebuilt pieces plus six that survived. Functionally, the recovered model scores 1.4665 on our sentences against the authors' 1.431 on their easier subset, and a one-index shift of the vocabulary scores above 5 bpc (as the checkpoint did before its state dict loaded). What is not available is an independent copy or the authors' token-id fixtures; the repaired files' hashes and an encoded fixture set are released so that anyone with either can check them.

Scoring. The authors' `score.py` with three patches that do not change the arithmetic (`weights_only=False`; the compiled-module prefix stripped and the state-dict load asserted; dynamic shapes for compilation); their protocol prefixes BOS, adds no end token, divides pooled sentence bits by pooled code points, and our sentences average 27 tokens in our tokenizer, far below the model's 1,024-token limit. The apostrophe check scored the first 2,000 sentences of the set with ASCII (1.3992) and typographic (1.4437) apostrophes.

I. Batch-size details

Per-seed values in D.3. Per-seed recoveries at mbs 8 under AdamW (seeds 47, 48, 49): wd 0.0056, 0.0073, 0.0121 (95% CI of the mean [−0.0000, +0.0167]); warmup −0.0020, −0.0002, 0.0054 ([−0.0085, +0.0107]); lr 0.0028, 0.0016, 0.0021; β 0.0109, 0.0164, 0.0092 ([+0.0028, +0.0215]); sum of singles 0.0173, 0.0251, 0.0288; all four 0.0273, 0.0258, 0.0300; joint minus sum of singles +0.0100, +0.0007, +0.0012, mean +0.0040 [−0.0090, +0.0170]. Because the single λ arm sits at λ = 0.05 and the joint arm at 0.0707, and the η arm also shortens the decay timescale, that residual mixes

interaction with a change of treatment level and is not an interaction estimate. All-four against the mbs-16 AdamW reference: −0.0033, −0.0024, −0.0028 (mean −0.0028 [−0.0040, −0.0017], t = 10.9). Timescale algebra: with batch B tokens per step, a horizon of S steps lasts B·S tokens, an EMA with retention β has token e-folding time −B/ln β, and the decoupled weight-decay timescale is B/(ηλ) tokens; halving B preserves each by doubling S, taking $\beta \to \sqrt{\beta}$ (0.9 → 0.949, 0.95 → 0.975) and halving ηλ (λ → 0.05 at fixed η in the single-knob arm; η and λ each × $1/\sqrt{2}$ in the joint arm). The joint arm's settings are those of the run table (`p1_adamw_batch`: lr 0.00361, wd 0.0707, warmup 400, β 0.95/0.975); an earlier draft omitted the λ value and was read as 0.05. The "~10%" learning-rate-mismatch estimate in §5.2 is from the curvatures of both groups under Muon+AdamW (AdamW 0.0007 + Muon 0.0022 ≈ 0.003 of 0.0275); the 9% share in Table 4 is the measured AdamW single-knob arm. Muon: against the masked mbs-8 reference (1.5294, three seeds) the halving penalty is +0.0299, paired by seed with the masked mbs-16 runs (0.0303, 0.0290, 0.0304; SE 0.0004; 95% CI [0.0280, 0.0318], df 2); the joint arm recovers 95% of it with the √B rule on both rates (ratio of means, delta method with the mbs-8 arm shared, SE ≈ 3 points, 95% CI 84–106%) and 91% (SE ≈ 1 point, 85–96%) with Muon's rate unscaled, ending +0.0014 [−0.002, +0.005] and +0.0028 [+0.001, +0.005] above the paired masked mbs-16 runs; the two Muon variants differ by 0.0014 (t = 1.2). The mbs-8 reference ran at AdamW 4.93e-3 (the full-validation vertex) and the mbs-16 runs at 5.10e-3 (the sweep point of Table 1); by the AdamW curvature that difference is worth < 0.0001. The frozen-settings penalty under Muon+AdamW exceeds AdamW's by 0.0050 (0.0299 vs 0.0249; combined SE ≈ 0.0013), so the orthogonalised group does not buy batch robustness at these settings; whether the excess is Muon's rate mismatch (0.0022 by the P1 curvature) or its momentum horizon is not separated by one joint arm.

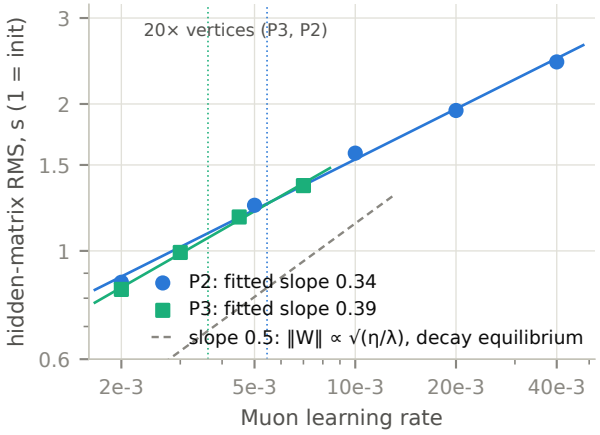


Figure I1. Hidden-matrix RMS at the end of training, relative to initialisation (s = RMS·√fan_in / 0.55), against the Muon learning rate, from the 20× sweeps at P2 and P3 (seed 47). Log–log slopes 0.34 (P2) and 0.39 (P3) against the 0.5 of the weight-decay equilibrium $\|W\| \propto \sqrt{(\eta/\lambda)}$: the weights sit near, not at, that equilibrium — which is why 1/(ηλ) is a horizon that a change of batch rescales (§5.2).

J. Muon overhead

19.9% of P1 step time in this implementation: five Newton–Schulz iterations on 24 matrices (fused QKV, output projection, fused gate–up, down projection, six layers; the fused matrices are orthogonalised jointly, which matters for anyone reproducing the learning-rate exponents), dispatch-bound, with about half of the overhead at P2 a fixed per-step cost independent of matrix size; ≈ 1% at the target. Only the wall-clock figure is quoted; a kernel-level profile accounted for 6.7% and does not close the gap.

K. Pre-registration record

Released with the code: the prediction files for each rung and the target with their timestamps and commit hashes; the frozen configuration file and the corpus, tokenizer and validation-set fingerprints; the evaluation code version for every stored number; the registered A, β and ray used for Table 2; and the deviations logged after registration (the 80×-tuned trunk rate; the instrument change; the padding fix). The 500M/1B gate of §8 was registered with the previous revision and is unchanged; the like-for-like forecast (0.883 / 0.839) is registered with this one. The 124M run had a

registered prediction but no registered gate, and the five-rung refit was a response to its result, not a pre-specified rule. Registered after the reviews and before the runs, with outcomes: (R1) masked small-batch Muon+AdamW reference at P1 — predicted halving penalty +0.024 to +0.030 and recovery ≥ 90%; measured +0.0299 (paired) and 95%. (R2) target at 20× at its own rate — predicted 0.999 ± 0.003 on val, −0.005 to −0.009 against the branch, and ≈ 1.10–1.12 on BrUK documents (a weak interpolation of the document ray); measured 1.0005, −0.0057 and 1.1186. (R2b) target at 20× from scratch at the branch's rate — predicted 1.0062 ± 0.002; measured 1.0061. Deviations logged: an attempt at R2 with Muon disabled by mistake (1.0316, retained as the AdamW-only point of §5.3); the packed-scorer fault of §7 and the re-score it required.

L. Specified runs and evaluations (status per item)

1. **Repetition curve at P2 [deferred to Part 2].** Train to 80× (786.8M tokens) at P2's 80× vertex (4.2e-3) on unique pools of 2,522M, 787M, 197M, 98M and 49M tokens ($\lambda \approx 0.31$, 1, 4, 8, 16 under the random-cut sampler), three paired seeds each, plus a fresh-data arm that samples the 2,522M pool *without replacement* (every token seen at most once; under random cuts the 787M pool at $\lambda = 1$ still leaves 37% of its tokens unseen and the 2,522M pool exposes only 676M unique tokens). Fit the effective-data half-life R^* across λ and report the $\lambda = 4$ penalty as a fitted value with an interval; Muennighoff et al. predict ≈ 0.002 ($\lambda = 4$), 0.005 (8) and 0.013 (16) bpc at P2, so the two larger repetition levels carry the fit. Pair nested pool draws and separate pool variation from training-seed variation. ≈ 8 GPU-hours. A result at P2 sizes the effect at P2; it does not by itself bound it at 124M.
2. **Matched-target context evaluation [eval].** One retained target set (the 32,960 filtered sentences, in document order with boundaries kept); score the same target positions with (a) no context, (b) the preceding sentences of the paragraph, (c) the preceding paragraphs of the document, with special-token targets excluded and the within-unit position $k \geq 8$ / 16 variants as a supplement. This replaces the protocol offsets of §6 (ii) with context effects on identical targets.
3. **Crossed-mask evaluation [eval].** Masked-trained and causal-trained checkpoints under both masks at every rung, plus a within-document text-only score; completes the one-sided junk test.
4. **Seed-bootstrap band for the ray; checkpoint-level instrument-difference table; per-seed branch-vs-scratch gates at P2–P4 [analysis/eval]; the target's gate is done (Appendix F).** Also feeds the gate band of §8.
5. **BrUK-document gate [eval].** 20× BrUK-document scores for the four rungs and the target; fit the ray and its leave-one-out range for §8.
6. **RoPE-reset identity check [eval].** Logit comparison of reset and no-reset positions on identical masked weights.
7. **Target at 20× at its own vertex [done 2026-10-05].** 2.49B tokens at Muon 0.85e-3, AdamW 1.19e-3, seed 47, full WSD: 1.0005 on val (registered 0.999 ± 0.003; paired −0.0057 against the branch, registered −0.005 to −0.009), 1.3617 on BrUK sentences. Folded into the abstract, §1, §4.3, Table 2, §4.4, §4.6, §6, Table 6, §8 and Appendix K.
8. **Ladder horizons at per-horizon rates [deferred to Part 2].** 40× and 80× trunks at their own vertices for P2–P4 (three seeds; vertices from the probes of Appendix E and a P4 probe), ≈ 19 GPU-hours; replaces Table 3's measured-as-detuned fits and the A(N) trend, and is the ladder Part 2 should inherit.
9. **Masked mbs-8 Muon+AdamW reference at P1 [done 2026-10-05].** Three seeds at the frozen configuration (run `paper1_final_r1`): 1.5282, 1.5301, 1.5300; paired penalty +0.0299, recovery 95% (√B) / 91% (unscaled); folded into §5.2, Appendix I and D.3.
10. **Verification of the tight masked arms and P2's β interval [done 2026-10-07].** Per-seed values from the run manifest (D.1): masked 80× seed sds 0.0008 (P2) and 0.0006 (P3) over five seeds; P2's per-seed β 0.599 / 0.608 / 0.591, SE 0.005. The run behind §7's worked example remains to be identified.

M. Supervised fine-tuning probe

Purpose. A qualitative check that the 124M checkpoint can be fine-tuned at all, and a shakedown of the fine-tuning instrument before Part 2 fine-tunes every rung with one frozen set and one recipe. Nothing here enters the pretraining claims.

Data. Four Ukrainian sets, each with a fixed validation split drawn from the full pool before training examples were sampled, frozen and fingerprinted: span extraction (UA-SQuAD from ua_datasets, MIT; 22,413 available; 7.1 answer tokens per example, 192 total), five-way news classification (UA-News from ua_datasets; 148,578; 2.3 / 304; class

shares 33.5 / 23.7 / 20.9 / 12.2 / 9.7%, so the majority baseline is 33.5%), grammatical error correction (UA-GEC 2.0, CC BY 4.0; 37,673; 21.1 / 50), and open responses (the Ukrainian portion of the Aya Dataset, Apache-2.0; 522; 96.8 / 157). Validation 500 / 500 / 500 / 200 = 1,700 examples; training 15,000 examples (an earlier 2,000-example set overfit by the third epoch).

Recipe. 8,192 tokens per step (micro-batch 4; ≈ 38 examples and ≈ 566 supervised tokens per step), three epochs, warmup 5%, linear decay to zero, learning rate = pretraining rate ÷ 10 with the same divisor at every rung, loss on response tokens and the end-of-answer token only, no loss across packing seams, best checkpoint by held-out loss. Token-weighted accumulation (Σ CE / supervised tokens); the end token's bits are amortised over the answer's characters when a bpc is reported. The chat template and special tokens were frozen before the first run.

What went wrong first, and the fixes. 1. *Generated BOS under document masking.* The block-diagonal mask treats BOS as a document start; a BOS the model samples therefore cuts attention to everything before it — the prompt becomes invisible, a hard cut rather than drift. Diagnosed from a missing leading-space marker on the first token after the cut (a document-initial token has none). Fix: set the BOS logit to $-\infty$ before sampling. Teacher-forced loss cannot see this fault; only generation can. 2. *Copy domination.* UA-GEC targets are ≈ 90% verbatim copies of their inputs. At 49% of the supervised tokens, the set put ≈ 44% of the total gradient on "repeat the input", which the model learned perfectly and which masked every other signal. Fix: copied spans masked out of the loss by a token-level diff of source and target (down-weighted rather than dropped, so the edits keep their context); the alternative, filtering the pool by normalised edit distance, biases toward broken sentences at higher thresholds. The general rule: with answer-only loss masking, the supervised-token *share* of each set, not its example count, sets its share of the gradient, and a rewriting set with near-identity targets is mostly copy supervision. 3. *Set size.* 2,000 → 15,000 examples.

Observations. - *First-token concentration.* The first answer token carries 55% of UA-GEC's held-out loss on 1/21 of its tokens; position 0 is where generation is decided. Up-weighting it did not help: held-out loss rose monotonically, 0.8630 → 0.8895 → 0.9002 → 0.9097 (trainer units, per answer token) at weights 1 / 3 / 5 / 10. Under autoregressive consistency the late-position gradient is already suppressed, so the learning signal concentrates at early tokens on its own — the mechanism the alignment literature calls shallow alignment (Qi et al. 2024). The remaining failure is a state-distribution mismatch: the model never sees an off-track state during teacher forcing, so it has no recovery policy (exposure bias; Ranzato et al. 2016). - *Outcome at 124M.* After the fixes: correct answer shapes, register mirroring, some facts right (Київ as capital; Коростень), many wrong (Рим for France), repetition loops present. Format acquired, knowledge absent — consistent with the model's size.

Carried into Part 2 (the capability curve across rungs): the same frozen set and recipe at every rung, three seeds, paired with the pretraining seeds; metrics reported per task and never aggregated — five-way accuracy against the 33.5% baseline, first-token accuracy, per-task held-out loss, and a 4-gram repetition rate; best-checkpoint selection per rung, since a fixed epoch count would measure memorisation speed rather than capability. Cost is ≈ 1% of pretraining tokens: all five models in minutes on one GPU.